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National Geospatial-Intelligence Agency

INSTRUCTION

NUMBER 8550.1
02 November 2020

CIO-T

SUBJECT: Health and Performance Monitoring of Information Systems and Applications

References: See Enclosure 1.

1. PURPOSE. This Instruction establishes policy and assigns responsibilities for monitoring the health and performance of NGA's information systems and applications (IS&As) in accordance with (IAW) References (b), (c), (d), and (g).

2. APPLICABILITY. This Instruction applies to all NGA civilian employees, military service members, other Government agency personnel assigned to NGA, and contractors, hereinafter referred to as *personnel*.

3. DEFINITIONS. See Glossary.

4. POLICY. It is NGA policy to monitor the health and performance of IS&As.

5. RESPONSIBILITIES. See Enclosure 2.

6. EFFECTIVE DATE. This Instruction is effective on the date of signature.

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Associate Chief Information Officer

Enclosures

1. References

2. Responsibilities

Glossary

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CHANGE LOG

Change	Change Date	Revised By	Pages Affected	Change Description

ENCLOSURE 1

REFERENCES

- (a) Committee on National Security Systems Instruction 4009, "National Information Assurance (IA) Glossary," 26 April 2010, revised 06 April 2015
- (b) DoD Instruction 8440.01, "DoD Information Technology Service Management," 24 December 2015
- (c) DoD "Enterprise Service Management Framework," Edition IV, 19 April 2019
- (d) NGA Directive (NGAD) 8000, "Information Management and the Chief Information Officer," 12 June 2013, Administrative Update 2, 28 October 2015
- (e) NGAD 8010, "Information Systems Risk Management Framework (RMF)," 27 August 2015, Administrative Update 1, 16 November 2015
- (f) NGA Instruction 3020.26, "Continuity of Operations Program," 17 May 2018
- (g) "NGA Enterprise Management Developer's Guide," Version 1.3, 8 June 2020 (U//FOUO) (located at <https://ngaonline/sites/CIO/T/TO/TOCME%20Docs/Forms/AllItems.aspx>)

ENCLOSURE 2

RESPONSIBILITIES

1. Chief Information Officer (CIO).

a. Oversees NGA information resources IAW Reference (d).

b. Delegates the responsibility to maintain and facilitate the implementation of a sound, integrated, and interoperable enterprise architecture for NGA to the Director, CIO and Information Technology (IT) Services (D/CIO-T) IAW Reference (d).

2. D/CIO-T.

a. Maintains and facilitates the implementation of a sound, integrated, and interoperable enterprise-wide architecture for NGA.

b. Assigns the responsibility to implement standardized enterprise-wide management (EM) IT health and performance monitoring of IS&As to Director, Secure Operations Group (D/TO).

3. D/TO.

a. Implements EM IT health and performance monitoring of IS&As.

b. Assigns the responsibility to maintain EM IT health and performance monitoring of IS&As to the Director, Operations Control Office (D/TOC).

c. Reviews and approves/disapproves waivers for existing and new IS&As with a Recovery Time Objective (RTO) of 0-72 hours to integrate IT health and performance monitoring tools into the Enterprise Service Center (ESC) Manager of Managers (MoM) and/or adopt enterprise management monitoring tools and reviews existing waivers semi-annually.

d. Encourages Program Managers (PMs) and IS Owners (ISOs) of IS&As with an RTO greater than 72 hours to integrate IT health and performance monitoring tools into the ESC MoM.

4. D/TOC.

a. Maintains EM IT health and performance monitoring of IS&As.

b. Maintains the NGA Enterprise Management Developers Guide (Reference (g)) and provides to PMs and ISOs to implement EM IT health and performance monitoring solutions for IS&As.

c. Ensures that PMs and ISOs are trained on EM IT health and performance monitoring tools.

d. Collaborates with PMs and ISOs to complete, unless granted a waiver, the required integration and adoption actions for existing IS&As with an RTO of 0-72 hours:

(1) Initiates integration of IT health and performance monitoring tools into the ESC MoM no later than (NLT) six months from the effective date of this Instruction.

(2) Completes integration of IT health and performance monitoring tools into the ESC MoM NLT 12 months from the effective date of this Instruction.

(3) Initiates adoption of EM IT health and performance monitoring tools NLT 18 months from the effective date of this Instruction.

(4) Completes adoption of EM IT health and performance monitoring tools NLT 01 October 2023.

(5) Develops, completes, and maintains a Service-Level Agreement (SLA) to identify and implement resolution and troubleshooting actions.

e. Approves SLAs and monitors the operational status of IS&As IAW approved SLA.

5. Directorate Heads and Agency Element Leads.

a. Ensure that all PMs and ISOs under their purview initiate and complete integration of IT health and performance monitoring tools IAW this Instruction.

b. Ensure that all PMs and ISOs under their purview initiate and complete adoption of EM IT health and performance monitoring tools IAW this Instruction.

6. PMs and ISOs.

a. Submit request for required integration and adoption actions in the Service+ tool.

b. Collaborate with D/TOC to complete, unless granted a waiver, the required integration and adoption actions for existing IS&As with an RTO of 0-72 hours:

(1) Initiate integration of IT health and performance monitoring tools into ESC MoM NLT six months from the effective date of this Instruction.

(2) Complete integration of IT health and performance monitoring tools into ESC MoM NLT 12 months from the effective date of this Instruction.

(3) Initiate adoption of EM IT health and performance monitoring tools NLT 18 months from the effective date of this Instruction.

(4) Complete adoption of EM IT health and performance monitoring tools NLT 01 October 2023.

(5) Develop, complete, and maintain the SLA to identify and implement resolution and troubleshooting actions.

c. Integrate IT health and performance monitoring tools for new IS&As with an RTO of 0-72 into the ESC MoM NLT 12 months after Initial Operational Capability (IOC).

d. Adopt EM IT health and performance monitoring for new IS&As with an RTO of 0-72 NLT 18 months after IOC.

e. Submit requests for integration or adoption actions for new IS&As with an RTO of 0-72 hours in the Service+ tool.

f. Submit request for a waiver of any requirement(s) in this Instruction to D/TO in the Service+ tool, and include justification for non-compliance such as cost, schedule, and/or technical limitation.

GLOSSARY

Application	A software program hosted by an information system. (Reference (a))
Enterprise	An organization with a defined mission/goal and a defined boundary, using information systems to execute that mission, and with responsibility for managing its own risks and performance. An enterprise may consist of all or some of the following business aspects: acquisition, program management, financial management (e.g., budgets), human resources, security, and information systems, information and mission management. (Reference (a))
Enterprise Architecture (EA)	A strategic information asset base that defines the mission, the information necessary to perform the mission, the technologies necessary for performing the mission, and the transitional process for implementing new technologies in response to changing mission needs. The EA includes a baseline architecture, target architecture, and sequencing plan. (Reference (a))
Enterprise Management (EM) Information Technology (IT) Health and Performance Monitoring	The process of monitoring and analyzing IT information system and applications performance data across the enterprise with the goal of improving IT service management and operational performance to meet business and mission needs. (Definition created by NGA for NGA Purposes Only)
Enterprise Service Center (ESC)	NGA's command and control center for Information Services and IT Infrastructure. It ensures end-to-end service delivery and performance for mission capabilities including imagery, imagery intelligence, and geospatial intelligence IT activities. (Definition created by NGA for NGA Purposes Only)
Information Resources	Information and related resources, such as personnel, equipment, funds, and information technology. (Reference (a))
Information System (IS)	A discrete set of information resources organized for the collection, processing, maintenance, use, sharing, dissemination, or disposition of information. Any equipment or interconnected system or subsystems of equipment that is used in the automatic acquisition, storage, manipulation, management, movement, control, display, switching, interchange, transmission, or reception of data or information, and includes computers and

	computer networks; ancillary equipment; software, firmware, and related procedures; services, including support services; and related resources. (Reference (d))
Information System Owner (ISO)	The individual responsible for the overall design, build, procurement, development, test, integration, modification, operation, maintenance, and disposal of an IS (as well as the system components) to include development and provision of the security plan. (Reference (e))
Information System and Application (IS&A) Performance Monitoring Tools	A suite of monitoring software comprising digital experience monitoring, application discovery, tracing, and diagnostics. IS&A performance monitoring tools provide insight and analytics into every part of the IS&A environment allowing for the ability to optimize application response times, transactions, and load times. (Definition created by NGA for NGA Purposes Only)
Initial Operational Capability (IOC)	A point in time during the production and deployment phase where a system can meet minimum operational capabilities for a user's stated need. (Definition created by NGA for NGA Purposes Only)
Manager of Managers (MoM)	An information system that collects and consolidates events from performance monitoring tools across all managed enclaves. The MoM provides role-based graphic views of Enterprise-wide systems and services for IT managers, operators, and technicians. (Definition created by NGA for NGA Purposes Only)
Program Manager (PM)	A system's government ISO. PM, as designated by the Directorate and/or Agency Element, is the government official responsible for the procurement, development, integration, modification, operation, maintenance, and disposal of an IS. (Reference (e))
Recovery Time Objective (RTO)	A value describing the maximum amount of time that a resource requirement or critical asset can be unavailable before it has a failure impact on essential functions. (Reference (f))
Service-Level Agreement (SLA)	A document that sets the expectations between the NGA Directorate and/or Agency Element and CIO-T. It describes the products or services to be delivered, the single point of contact for end-user problems, and the metrics by which the effectiveness of the process is monitored and approved. (Definition created by NGA for NGA Purposes Only)

Service+ tool A management tool that combines and automates key business processes to keep NGA's infrastructure operating properly. Generates action requests to track events such as customer service requests. Provides a user friendly graphical user interface and consolidates action tracking information into a single database allowing for access to the information based upon user rights and privileges. (Definition created by NGA for NGA's purposes only)